（a）Data foundation
（b）Engineering modeling
（c）QAE–ABC optimization
North Sea
Historical North Sea metocean data
North Sea reanalysis
Historical environmental samples
Fitting data
Out-of-time data
OC4 structural schematic
Fitting-period records
Low-fidelity Analysis
OpenFAST Simulations
Multi-seed 600 s OpenFAST simulations
Multifidelity Response Surrogate
Response-specific regression
Surrogate–OpenFAST comparison
Reliability Metamodel
Probability prediction
QAE amplitude simulation
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|0⟩
Amplitude simulation
Interval Ranking Design Update
ABC candidate
min M(d) s.t. P(d) ≤ p